

**PROJECT CONCEPT PROPOSAL***(To be accomplished by the researcher)*

Dear Proponent:

Thank you for applying to the DOST-GIA program. We value your proposal and your time for accomplishing the form. Approved projects are the government's investments on you and your institution. Your answers are very important to us as they may indicate your potential contributions to these public investments in research and development. We wish you all the best.

Sincerely,
DOST

I. CONCEPT PROJECT PROPOSAL SUMMARY INFORMATION

1. Project Title	<i>Resilience Assessment of Remaining Swine Population as Affected by African Swine Fever Virus in Panay Island</i>
2. Project Duration <i>(in months)</i>	24 months
3. Estimated Budgetary Requirement	
4. Principal Research Question to be Addressed <i>(maximum of 100 words)</i>	

II. PROJECT LEADER PROFILE

1) Name	2) Position	3) Organization	4) Department/ Division	5) Contact Details <i>(telephone number, email)</i>
6) Other Ongoing Projects Being Handled by the Project Leader: _____ (number)				
Title of the Project		Funding Agency		Involvement in the Project

III. PROJECT PROFILE

1. Cooperating Agency/ies <i>(indicate role/responsibility of cooperating agency)</i>			
2. Site (s) of Implementation <i>(Municipality/District/Province/ Region)</i>	Base Station:		
	Other implementation sites:		
3. R&D Program		Basic	Applied
a) Agriculture, Aquatic and Natural Resources			X
b) Health			
c) Industry, Energy and Emerging Technology			

d)	Disaster Risk Reduction and Climate Change Adaptation		
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4. DOST Pillars Pursued <i>(Check all that apply)</i> ☐ Human Wellbeing ☐ Wealth Creation ☐ Wealth Protection ☒ Sustainability	(5) DOST Thematic Areas Covered <i>(Check all that apply)</i> ☐ Learning, Education, and Culture ☐ Health and Wellbeing ☒ Food and Agriculture ☐ Industrial Solutions and Competitiveness Development ☐ Transport and Mobility ☐ Energy and Utilities System ☐ Environment & Natural Resources ☐ Climate, Disaster Resilience, and Human Security ☐ Governance ☐ Others (please specify): _____
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(6) Project Summary

(Please briefly describe your proposed research project ensuring the clarity of the scientific rationale; the deficit, needs, gaps or problems to be addressed; significance of the research and the solutions/benefits that it will offer; and relevance to the Philippine context. Include basis of problem identification (i.e., roadmap, value chain, industry request, etc.). Explicitly identify the objectives of the project and its target accomplishments and outputs. (maximum of 1,500 words).

The African swine fever (ASF) is a severe, highly contagious viral disease that currently has no vaccine that affects both domestic pigs and wild boars. In the Philippines, the first reported case of ASF was in July 2019 in Rizal Province, wiping out over a third of the country's pig stocks valued at 200 billion pesos, threatening food security in a country already reeling from the economic impact due to COVID19 pandemic. This has resulted in severe pork shortages, an important part of the Filipino diet, which could lead to a rise in the prices of many food items. Hence, the growth rate of pork production dramatically decreased by 20.8% in 2021, impacting vulnerable consumers' livelihoods.

As part of the government's initiative to recover the swine industry's capacity to sufficiently produce hogs and pork by the year 2023, it allocated Php 4.1 Billion under GAA 2022 for the implementation of the Integrated National Swine Production Initiatives (INSPIRE) and Bantay ASF sa Barangay or BABay ASF Programs, which aims to institutionalize biosecurity and disease control measures in tandem with calibrated repopulation and enhancement of the local swine industry.

Nevertheless, challenges in its implementation are inevitable. Insufficient workforce from the LGU focusing on the recovery protocol activities plays as a major hindrance to the execution of a science-based approach to proceed with the repopulation. This information is critical for the succeeding activities towards the attainment of the goal of the DA to sustain the swine industry. Part of the protocol are culling of infected and exposed pigs, followed by cleaning and disinfection, environmental surveillance of affected areas and placement of sentinel animals.

To date, data on the environmental surveillance which serves as the baseline for the zone progression and local declaration of ASF-Free Status is insufficient. There is a need to augment workforce and expertise from other agencies such as DA-ATI, SUCs, Genome Center and Livestock Biotech Center to come up with a comprehensive information on the remaining swine population and their resilience to ASF.

Profiling of swine farms and biobanking of the remaining stocks will be done simultaneous to GIS Mapping activities to highlight the best practices of swine farmers in the prevention of ASF, collect, process and store biological samples and data for further investigations, preventive measures and vaccine development for ASF. GIS mapping will be done to assess the trend in the prevalence and pathogenicity of ASF, and provide a visual representation of the outbreak.

Region 6 (Western Visayas) ranks 2nd swine producer in the country contributing 1,166,400 heads of hogs as of July 1, 2022 (Panay News 2022). However, as of September 30, 2023 Western Visayas abated to number 6 among the country's top producing regions (PSA 2023).

This abatement on the volume of production needs to have an inclusive investigation to evaluate the differences in the practices among swine raisers, understand patterns and relationship among ASF infected areas, and assess the expression of immune response likewise the biological profile of the remaining hogs in Region 6 hence this study.

7)	SUBMITTED BY (Project Leader)
Signature	
Printed Name	EMELY J. ESCALA, PhD
Designation/Title	Director, Livestock Research and Development Center
Date	2/23/2024

Note: If the project is part of a program, this form should be submitted together with the program concept proposal and one (1) curriculum vitae. Use A4 size paper, Arial typeface 10 points.